```r
# ============================================
# Program: base_pop_data_vis.R
# Author: Jameson Roth
# Date: 2026 10 27
# Description: Data visualisations and summary stats of base population (2011 zips)
# ============================================

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------------------------------------------
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------------------------------------------------
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(rprojroot)

setwd(find_root(is_git_root))

pop <- read_csv("0_data/1_clean/base_population.csv") |>
    group_by(cbsa) |>
    mutate(
        count_gentrifiable = sum(gentrifiable, na.rm = TRUE),
        total_zips = n(),
        share_gentrifiable = count_gentrifiable / total_zips,
        cbsa = as.character(cbsa),
        cbsa_name = case_when(
            cbsa == "14460" ~ "Boston",
            cbsa == "47900" ~ "Washington D.C.",
            cbsa == "31080" ~ "Los Angeles",
            cbsa == "41860" ~ "San Francisco",
            cbsa == "35620" ~ "New York",
            cbsa == "16980" ~ "Chicago",
        )
    ) |>
    ungroup() |>
    select(cbsa, count_gentrifiable, share_gentrifiable, cbsa_name) |>
    unique()
```

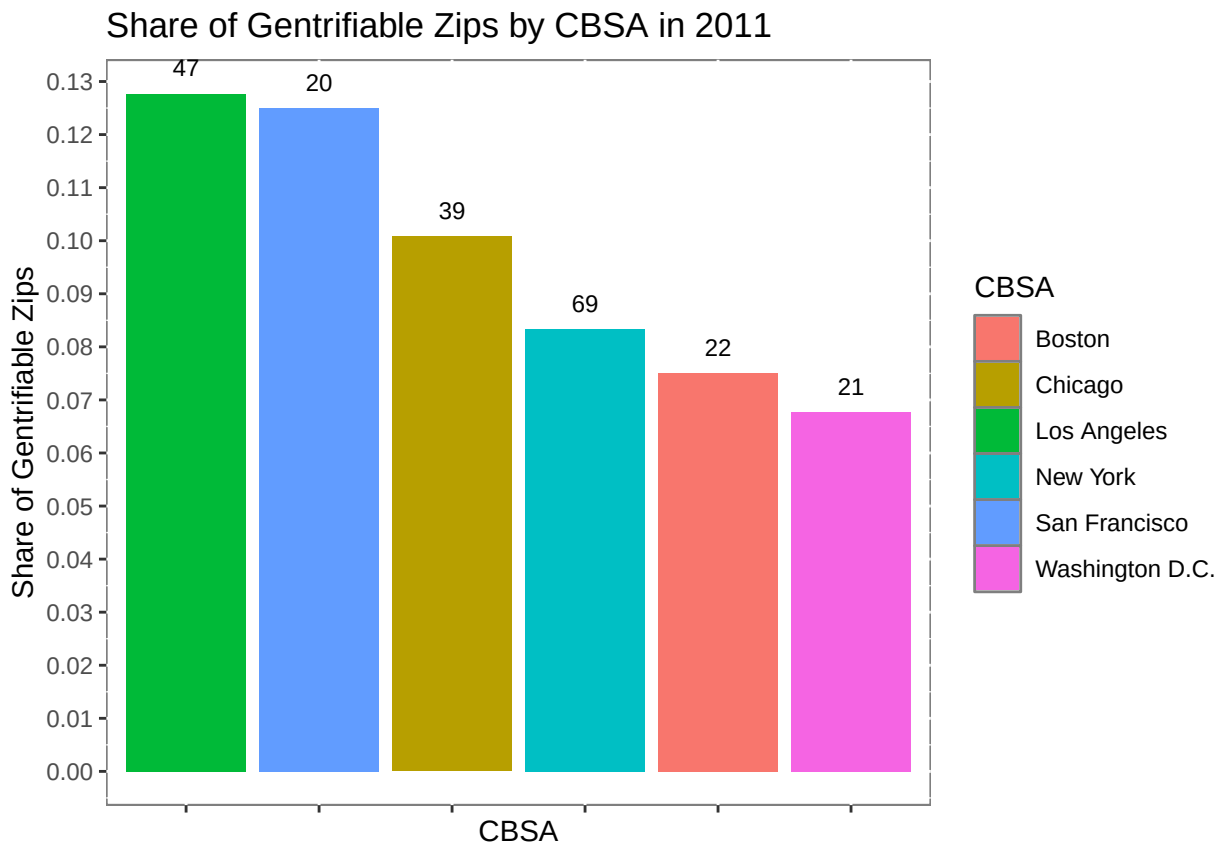
```
## Rows: 2346 Columns: 10
## -- Column specification --------------------------------------------------------------------------
## Delimiter: ","
## chr (1): zipcode
## dbl (8): statefips, year, cbsa, zip_median, cbsa_40pct, annual_avg_zhvi, med...
## lgl (1): gentrifiable
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(pop)
```

```
## Rows: 6
## Columns: 4
## $ cbsa               <chr> "31080", "41860", "47900", "16980", "14460", "35620"
## $ count_gentrifiable <int> 47, 20, 21, 39, 22, 69
## $ share_gentrifiable <dbl> 0.12771739, 0.12500000, 0.06774194, 0.10077519, 0.0~
## $ cbsa_name          <chr> "Los Angeles", "San Francisco", "Washington D.C.", ~
```

```r
# Share gentrifiable by CBSA
chart <- ggplot(
    pop,
    aes(
        x = reorder(cbsa_name, -share_gentrifiable),
        y = share_gentrifiable,
        fill = cbsa_name,
    )
) +
    geom_text(aes(label = count_gentrifiable), vjust=-1, size=3) +
    geom_bar(stat = "identity") +
    xlab("CBSA") +
    ylab("Share of Gentrifiable Zips") +
    scale_y_continuous(breaks = seq(0, 0.5, 0.01)) +
    scale_fill_discrete(name = "CBSA") +
    ggtitle("Share of Gentrifiable Zips by CBSA in 2011") +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
    )

show(chart)
```


Share of Gentrifiable Zips by CBSA in 2011
Share of Gentrifiable Zips
0.13
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
47
20
39
69
22
21
CBSA
CBSA
Boston
Chicago
Los Angeles
New York
San Francisco
Washington D.C.